

Name: N. Sai Sreenu

Reg No: 192365001

C05-AT1

MLA0201-Fundamentals Of machine Learning

Case Study Answer Sheet

Question 1: Reinforcement Learning for Warehouse Robot Navigation (20 Marks)

1. Problem Understanding

The warehouse robot navigation task is formulated as a Markov Decision Process (MDP), defined by the tuple (S, A, P, R, γ) . The robot must travel from a start location to a dispatch (goal) location on a grid-based warehouse layout while avoiding fixed obstacles such as storage racks. Because the environment is stochastic, an action chosen by the robot does not always result in the intended movement; there is a probability that the robot drifts to an unintended adjacent cell. The problem therefore requires a policy that is optimal not for a single deterministic path, but in expectation over all possible stochastic outcomes.

The MDP components are summarised below.

<u>MDP Element</u>	<u>Description in this Scenario</u>
State (S)	Each grid cell (x, y) that the robot can occupy, including battery level as part of the state.
Action (A)	Four discrete actions: Up, Down, Left, Right.
Transition $P(s' s,a)$	Stochastic; the intended action succeeds with high probability, with a residual probability of slipping to a neighbouring cell.
Reward (R)	Positive reward on reaching the dispatch area, small negative reward (step cost) per move, large negative reward for colliding with an obstacle.
Terminal condition	Destination reached, battery exhausted (30 units), or move limit exceeded (25 moves).

2. Constraint Analysis

The constraints given in the scenario are analysed and prioritised below, from the most operationally critical to the least.

<u>Priority</u>	<u>Constraint</u>	<u>Impact on the RL Formulation</u>
High	Battery capacity of 30 units	Modelled as a depleting resource; each move reduces battery by one unit, so the policy must reach the goal before the resource is exhausted. This is encoded as part of the state and as a step-cost in the reward function.
High	Obstacles cannot be crossed	Obstacle cells are removed from the valid action space; attempting to enter one yields a heavy penalty and the robot remains in place.
Medium	25-move limit	Acts as a finite time horizon; encourages the agent to prefer shorter paths through the discount factor γ and the per-step penalty.
Medium	Stochastic transitions	Requires the policy to be robust to unintended outcomes; Value Iteration explicitly incorporates transition probabilities in the Bellman equation.
Low	Reward and penalty structure	Shapes the relative preference between short, safe paths and longer but risk-free paths.

3. Solution Development

Exploration Strategy: Because the transition probabilities of the stochastic environment are not fully known in advance, an ϵ -greedy exploration strategy with a decaying ϵ is recommended during the learning phase in which the robot interacts with the environment (or a simulator) to estimate the transition model. Early episodes use a higher ϵ (e.g., 0.3) to encourage exploration of the grid, and ϵ is annealed towards a low value (e.g., 0.05) as the robot gathers sufficient experience, so that exploitation of the learned model dominates once the estimates stabilise. This prevents the robot from prematurely converging on a suboptimal or unsafe path.

Planning Algorithm: Once the transition probabilities and rewards are estimated (or if a model of the warehouse is available upfront), Value Iteration is applied to compute the optimal policy. Value Iteration repeatedly applies the Bellman optimality update:

$$V(s) \leftarrow \max_{\mathbf{a}} [R(s,\mathbf{a}) + \gamma \sum P(s'|s,\mathbf{a}) V(s')]$$

until the value function converges, after which the optimal policy is extracted greedily from the converged values. The battery and move-count constraints are incorporated by augmenting the state with remaining battery/steps, so that the value function naturally discourages paths that would exhaust the battery or exceed 25 moves before reaching the goal.

4. Application of Concepts

The solution applies core RL concepts as follows: (i) the Bellman optimality equation underlies the Value Iteration updates; (ii) the discount factor γ (recommended 0.9–0.95) balances immediate step cost against the long-term goal reward; (iii) reward shaping (small negative step reward, large terminal reward, obstacle penalty) directs the agent towards battery-efficient, obstacle-free paths; and (iv) the ϵ -greedy strategy supplies the exploration needed to learn accurate transition estimates in the stochastic environment before planning is performed.

5. Justification

Value Iteration is preferred over Policy Iteration for this scenario because the state space (grid cells combined with battery and step counters) is moderate in size and Value Iteration converges directly on the optimal value function without the repeated, computationally heavier policy-evaluation sweeps that Policy Iteration requires at every iteration. Value Iteration also adapts naturally to the stochastic transition model because each update already integrates over all possible next states. ϵ -greedy exploration is justified because it is simple to implement, allows explicit control over the exploration rate through decay, and is well suited to a discrete, moderate-sized action space such as the four-directional movement used here.

Question 2: Multi-Armed Bandit for Online Advertisement Selection (20 Marks)

1. Problem Understanding

This problem is a classic K-Armed Bandit problem. Each of the five advertisements represents one "arm" of the bandit. Displaying an advertisement to a visitor is equivalent to pulling an arm, and a click (reward = 1) or no click (reward = 0) is the observed outcome. The true click-through probability of each advertisement is unknown at the start and must be estimated online, using the visitor interactions themselves, while simultaneously trying to maximise the total number of clicks obtained over the 10,000-interaction budget.

<u>Bandit Element</u>	<u>Mapping to the Scenario</u>
Arms ($K = 5$)	The five candidate advertisements.
Action	Choosing which advertisement to display to the current visitor.
Reward	Binary: 1 if the visitor clicks the advertisement, 0 otherwise (Bernoulli reward).
Unknown parameter	The true click-through probability p_i of each advertisement i .
Budget / horizon	10,000 total visitor interactions.
Objective	Maximise cumulative clicks, equivalently minimise cumulative regret relative to always showing the best advertisement.

2. Constraint Analysis

<u>Priority</u>	<u>Constraint</u>	<u>Implication</u>
High	Only one advertisement shown per visitor	The algorithm must make a single, immediate decision per interaction, ruling out any strategy that requires showing several ads to the same visitor.
High	Unknown click probabilities	Requires an online estimation mechanism (e.g., running average or Bayesian posterior) that is updated after every interaction.
Medium	Fixed budget of 10,000 interactions	Limits the total exploration that can be afforded; the exploration strategy must converge quickly to avoid wasting the budget on clearly inferior advertisements.
Medium	Minimise cumulative regret	Favours strategies with theoretically bounded regret growth, such as UCB or Thompson Sampling, over strategies with constant exploration, such as fixed ϵ -greedy.

3. Solution Development

Thompson Sampling is recommended as the exploration strategy. Each advertisement i is modelled with a $\text{Beta}(\alpha_i, \beta_i)$ posterior distribution over its unknown click probability, initialised as $\text{Beta}(1,1)$ (a uniform prior) for all five advertisements. For every visitor: (1) a sample is drawn from each advertisement's current posterior distribution, (2) the advertisement with the highest sampled value is displayed, and (3) the corresponding posterior is updated using the observed outcome — α_i is incremented by 1 on a click and β_i is incremented by 1 otherwise. This procedure is repeated for all 10,000 interactions.

As an alternative, Upper Confidence Bound (UCB1) may be used, selecting the advertisement that maximises the estimated click-through rate plus a confidence bonus that shrinks as an advertisement is shown more often. Both approaches are preferable to a fixed ϵ -greedy policy, which continues to explore inferior advertisements at a constant rate for the entire budget.

4. Application of Concepts

The solution applies the exploration–exploitation trade-off central to bandit theory: exploration is required because click probabilities are unknown, while exploitation is required to make the most of the limited budget. Thompson Sampling applies Bayesian updating of Beta-Bernoulli conjugate

distributions, naturally reducing exploration of an arm as evidence accumulates. Cumulative regret, defined as the gap between the reward obtained from the optimal arm and the reward actually obtained, is used as the formal performance metric that both Thompson Sampling and UCB are designed to minimise.

5. Justification

Thompson Sampling is justified for this scenario because click/no-click outcomes are naturally Bernoulli, for which the Beta distribution is the conjugate prior, allowing efficient closed-form posterior updates. Empirically and theoretically, Thompson Sampling achieves regret comparable to or better than UCB in many practical advertisement-selection settings, and it explores advertisements in proportion to the current uncertainty about their performance rather than through a fixed schedule, which makes efficient use of the constrained 10,000-interaction budget. As more data accumulates for a given advertisement, its posterior narrows and the algorithm is exploited more heavily if it performs well, directly reducing cumulative regret while maximising total clicks.

Reward mechanism: because each click directly updates the posterior in favour of the advertisement just shown, advertisements that perform well are selected with increasing frequency over time, while advertisements with low or zero observed click rates are shown progressively less often, letting the reward signal itself drive the shift from exploration towards exploitation.

Question 3: Sampling Strategy for Machine Learning Model Development (20 Marks)

1. Problem Understanding

A disease-prediction model must be trained on a dataset of one million patient records, but the available training memory (8 GB RAM) makes it infeasible to load and train on the complete dataset directly. A representative subset must therefore be sampled such that the resulting model still achieves at least 95% accuracy with at least 95% confidence, while keeping sampling bias and computational cost low. The core challenge is choosing a sampling method that preserves the statistical structure of all patient categories (e.g., disease subtypes, age groups, demographics) present in the full population.

2. Constraint Analysis

<u>Priority</u>	<u>Constraint</u>	<u>Implication for the Sampling Strategy</u>
High	8 GB RAM limit	Bounds the maximum sample size that can be held in memory during training, requiring a sample considerably smaller than one million records.

<u>Priority</u>	<u>Constraint</u>	<u>Implication for the Sampling Strategy</u>
High	Sample must represent all patient categories	Rules out plain Simple Random Sampling in cases of class imbalance, since rare categories could be under-represented or missed entirely.
High	$\geq 95\%$ accuracy and $\geq 95\%$ confidence	Requires the sample size to be computed from statistical sample-complexity bounds relating accuracy (ϵ) and confidence ($1 - \delta$) rather than chosen arbitrarily.
Medium	Minimise sampling bias and computational cost	Favours a structured sampling method over exhaustive or purely random approaches.

3. Solution Development

Stratified Sampling is recommended. The one million patient records are first divided into homogeneous strata based on relevant categorical attributes (e.g., disease status, age band, gender, comorbidity group). A sample is then drawn from each stratum in proportion to its size in the overall population. This guarantees that every patient category is represented in the training sample in roughly the same proportion as in the full dataset, directly satisfying the requirement to represent all patient categories and to minimise sampling bias, while keeping the resulting sample small enough to fit within the 8 GB memory budget.

The required sample size n is estimated using standard sample-complexity bounds for a desired accuracy ϵ and confidence level ($1 - \delta$):

$$n \geq (1 / (2\epsilon^2)) \cdot \ln(2 / \delta)$$

For ϵ corresponding to 95% accuracy and $\delta = 0.05$ (95% confidence), this bound yields a concrete target sample size that is orders of magnitude smaller than one million records, making the 8 GB constraint achievable. Monte Carlo resampling (e.g., bootstrapping) can additionally be used within each stratum to validate that the estimated accuracy is stable across repeated draws.

4. Application of Concepts

<u>Concept</u>	<u>Role in This Scenario</u>
Sample Complexity	Determines the minimum number of records n needed so that, with probability at least $1 - \delta$, the trained model's error is within ϵ of the true error, directly linking the 95% accuracy and 95% confidence requirements to a concrete sample size.
VC Dimension	Bounds sample complexity as a function of hypothesis class complexity; a hypothesis class with higher VC dimension requires

<u>Concept</u>	<u>Role in This Scenario</u>
	proportionally more samples to guarantee the same accuracy/confidence, so the model family chosen should have VC dimension appropriate to the stratified sample size available.
Occam's Learning Principle	Favours the simplest hypothesis consistent with the training data, reducing the risk of overfitting to the sampled subset and improving generalisation to the full one-million-record population.
Accuracy–Confidence Boosting	Ensemble techniques (e.g., boosting) combine multiple weak learners trained on the stratified sample to raise overall accuracy while tightening the achievable confidence bound, compensating for the reduced sample size relative to the full dataset.

5. Justification

Stratified Sampling is justified over Simple Random Sampling because random sampling alone cannot guarantee proportional representation of minority patient categories, which is critical in healthcare data where rare disease subtypes may otherwise be missed, biasing the model. It is justified over pure Monte Carlo sampling because Monte Carlo methods are better suited to simulation-based estimation than to constructing a single fixed training set with guaranteed category coverage; however, Monte Carlo resampling is retained as a complementary validation step. Combined with the sample-complexity bound, VC-dimension-aware model selection, Occam's principle, and boosting, the stratified approach satisfies the memory constraint, the representativeness requirement, and the 95% accuracy/confidence targets simultaneously, while minimising both bias and computational cost.

Rubric Reference

The following rubric was used as a reference to structure and weight each answer.

<u>Criteria</u>	<u>Weight</u>	<u>Level 4 (Exemplary)</u>
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately.
Constraint Analysis	25%	Analyses constraints effectively and prioritises them logically.
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints.

<u>Criteria</u>	<u>Weight</u>	<u>Level 4 (Exemplary)</u>
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools.
Justification	10%	Provides strong justification for decisions with logical reasoning.